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SKILL WALLET
A SmartBridge Product

Project Proposal (Proposed Solution)

Date	28 June 2026
Team ID	
Project Name	Rising Waters-ML Flood Prediction System
Maximum Marks	5 Marks

Project Proposal(Proposed Solution) Report

Project Overview	
Objective	To develop an intelligent flood prediction system using Machine Learning that analyzes rainfall and weather parameters to accurately predict flood risk. The system aims to support disaster preparedness by providing early flood warnings through a user-friendly Flask web application.
Scope	The project covers the complete machine learning pipeline, including dataset collection, data visualization, preprocessing, model training, evaluation, and deployment. It enables users to input weather parameters such as annual rainfall, seasonal rainfall, cloud visibility, temperature, and humidity to obtain real-time flood prediction results. The system is intended for disaster management authorities, meteorological departments, researchers, and communities in flood-prone areas.
Problem Statement	
Description	Floods are among the most destructive natural disasters, causing significant loss of life, property, and infrastructure every year. Traditional flood forecasting methods often require extensive manual analysis and may not provide timely predictions. There is a need for an intelligent system that can quickly analyze weather data and accurately predict flood risk to support early warning and disaster management.
Impact	● Improves disaster preparedness and response. ● Reduces the risk of loss of life and property. ● Supports government and disaster management authorities in decision-making. ● Provides quick and reliable flood predictions. ● Encourages the adoption of Artificial Intelligence in environmental monitoring.
Proposed Solution	

Approach	The proposed solution uses Python-based Machine Learning techniques to predict flood risk from weather and rainfall data. The project begins with collecting a flood prediction dataset from Kaggle, followed by exploratory data analysis and preprocessing. Multiple classification algorithms, including Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost, are trained and evaluated. The XGBoost model is selected based on its superior accuracy and saved for deployment. A Flask web application allows users to enter weather parameters and instantly receive flood prediction results.
Key Features	● User-friendly Flask web application. ● Weather parameter input interface. ● Machine Learning-based flood prediction. ● XGBoost model for high prediction accuracy. ● Data visualization and statistical analysis. ● Data preprocessing and feature scaling. ● Instant prediction of flood or no-flood conditions. ● Prediction results displayed with minimal processing time. ● Model ready for IBM Cloud deployment.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel Core i3 or above
Memory	RAM	Minimum 4 GB
Storage	Disk Space	Minimum 2 GB free storage
Software		
Frameworks	Web Framework	Flask
Libraries	Python Libraries	NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Joblib/Pickle, XGBoost
Development Environment	IDE & Platform	Python 3.8+, Anaconda Navigator,Jupyter Notebook, Windows/Linux/macOS, IBM Cloud Account
Data		
Data	Training Dataset	Flood prediction dataset containing annual rainfall, seasonal rainfall, cloud visibility, temperature, humidity, and other meteorological parameters collected from Kaggle.